Equation of a tangent from a graph

1

a

i $(1, 0)$ ii 2 iii $y = 2x - 2$

b

i $(1, -3)$ ii -2 iii $y = -2x - 1$

c

i $(2, 2)$ ii -1 iii $y = -x + 4$

d

i $(4, 0)$ ii 1 iii $y = x - 4$

e

i $(2, 3)$ ii $\frac{1}{2}$ iii $y = \frac{x}{2} + 2$

2

a

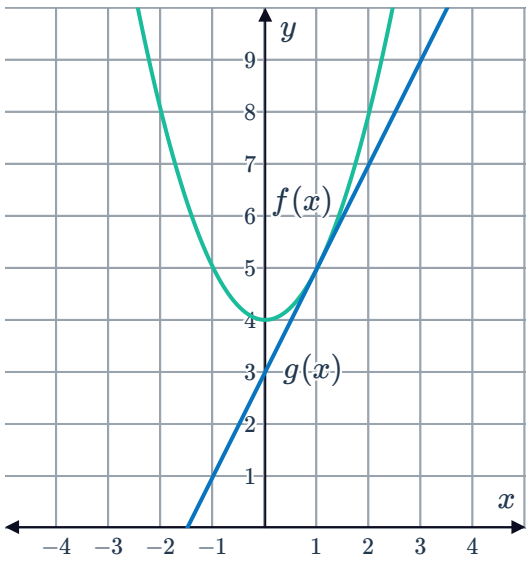
i $(-1, -6)$ ii 3 iii $y = 3x - 3$ iv $x = 1$

b

i $(1, -4)$ ii -3 iii $y = -3x - 1$ iv $x = -1$

3

a

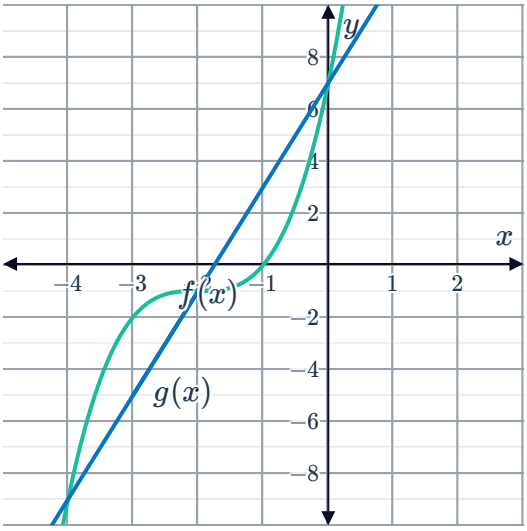


b Yes, because the line $g(x)$ touches the curve $f(x)$.

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6

a

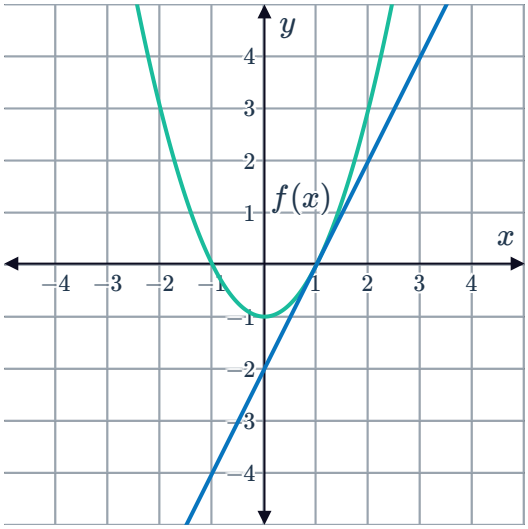


b No, because the line $g\,(x)$ crosses the curve $f\,(x)$ three times

7

a 2

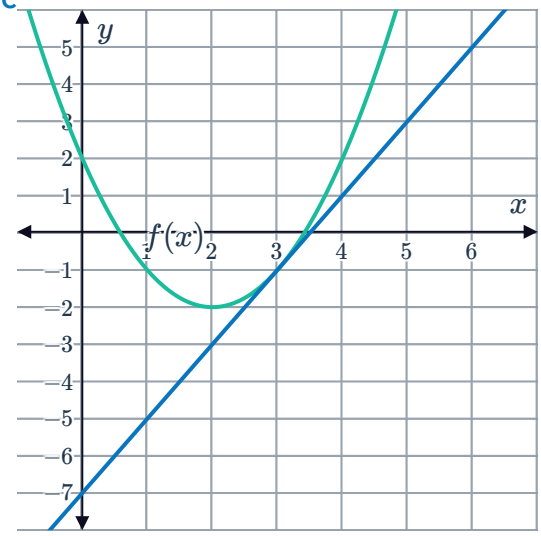
b



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8

a 2 **b** $(2, -2)$ **c**



Equation of a tangent at a given point

9 $f'(2) = 17$

10

a $x = -5, x = 2$ **b** $f'(2) = 7$

11 $x = 2$

12

a $\dfrac{dy}{dx} = x^2 + x^5 + 2$ **b** $\dfrac{dy}{dx} = -16\,756$

13

a $f'(-1) = -2$ **b** $y = -2x - 1$

14

a $f'(-1) = 3$ **b** $y = 3x + 2$

15

a $f'(-3) = 6$ **b** $y = 6x + 9$

16

a $f'(2) = -12$ **b** $y = -12x + 16$

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17

a $f'(-2) = -4$ **b** $y = -4x - 4$

18

a $f'(4) = \frac{3}{2}$ **b** $y = \frac{3x}{2} + 6$

19

a $f'\left(\frac{1}{9}\right) = \frac{15}{2}$ **b** $y = \frac{15x}{2} + \frac{5}{6}$

20

a $y = 7x - 19$ **b** $y = 5x - 4$ **c** $y = 5x - 16$ **d** $y = 2x - 9$

e $y = -18x + 27$

21

a To find the equation of the tangent at $x = 2$, we need to find the gradient of the tangent at that point, which is given by $f'(2)$. Then we need to find the y-coordinate of the point at $x = 2$. Then we can substitute the point and the gradient into the point-gradient formula $y - y_1 = m(x - x_1)$

b $y = 36x - 48$

22

a $f'(-2) = -\frac{3}{8}$ **b** $y = -\frac{1}{4}$ **c** $y = -\frac{3x}{8} - 1$

23

a $f'(2) = 12$ **b** $y = 12$ **c** $y = 12x - 12$

24

a $y = 8$ **b** $y = 4x + 4$

25 $y = -10.1x + 8.4$

26 $y = -\frac{4x}{3} + \frac{1}{3}$

27 $y = 7x - \frac{1}{2}$

28

a $m = 8, 4$ **b** $y = 8x - 22, y = 4x - 10$

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29 $y = -13$

30

a $x = \frac{1}{2}, \frac{1}{6}$

b $y = -\frac{1}{4}x,\ y = -\frac{x}{4} + \frac{1}{54}$

Applications

31

a

i $x = 2$ **ii** $y = 4$

b

i $x = 2$ **ii** $y = 8$

32 $x = \frac{3}{4}, x = -2$

33

a $\frac{dy}{dx} = 2x + b$ **b** -5 **c** $b = -23$ **d** $c = 79$

34

a $x = 4$

b $(4, 36)$

35

a $x = 15, -11$

b $(15, 2025)\,,\ (-11, -2057)$

36

a $x = 2, -4$

b $(2, -16)\,, (-4, 62)$

37

a $x = 2, x = -6$

b $(2, 2)\,, (-6, 82)$

38 $k = 40$

39

a $\frac{dy}{dx} = 8x - 5$

b $x = \frac{5}{8}$

40 $\theta = 87^\circ$

41 $a = 3, b = 5$

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42 $a = 2, b = 3, c = 0, d = -28$

43

a $a = 900$

b $b = 90$

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